

Create Dynamic Graphics in PHP

In this article, the authors take the reader all the way through the process of creating dynamic graphics, right from setting up the Graphics Drawing library to drawing various types of graphs.



Nowadays, dynamic images, graphical reports, charts of different types, maps and graphs are used in Web applications so that effective and real-time results can be displayed and exported with effective graphics.

All server side scripts integrate graphics and charting libraries to include dynamic images in multiple customised formats. SVG (Scalable Vector Graphics) is one of the classical image formats that has been integrated with Web applications but there are many limitations. In SVG, it is difficult to create dynamic graphs and charts of different types as is required in many Web applications. All major HTTP clients or Web browsers, including Mozilla Firefox, Safari, Internet Explorer, Google Chrome and Opera, integrate SVG rendering support. SVG is basically an XML-based vector image format that is used for two-dimensional graphics with support for interactivity and animation. The SVG specification is an open standard that has been developed by the World Wide Web Consortium (W3C) since 1999. Using SVG is not advised if you need to integrate pie charts, bar charts, maps or dynamic identity cards, online certificates or similar imaging features. Neither is it used for 3-D rendering,

for which alternate technologies are preferred.

GD library in PHP

The URLs for the library are:

- <http://libgd.bitbucket.org>
- <http://www.boutell.com/gd/>

GD library is the graphics drawing library that integrates the tools for creating and manipulating image data. GD is used with PHP to create and render the dynamic images in multiple formats including GIF, PNG, JPEG, WBMP, XPM and WebP. Using PHP GD, the output image streams can be directly displayed to a Web browser. PHP GD requires some other libraries, depending on which image formats we want to work with. We can use the image functions in PHP to get the size of JPEG, GIF, PNG, SWF, TIFF and JPEG2000 images.

Using the *exif* extension, we can work with the information that is stored in the headers of JPEG and TIFF images. In this way, we can read and process the meta data generated by digital cameras. GD library enables the programmer to create the dynamic charts, anti-robot security images (CAPTCHA), thumbnail images or even hybrid